

Hi!

- I am Gabe Chodorow-Reich, your instructor for the remainder of the course.
- Harvard '05 (Social Studies, Adams House), 1011B '03.
- What sort of research do I do?
 - ▶ <https://scholar.harvard.edu/chodorow-reich/publications>.
 - ▶ <https://www.nytimes.com/2024/03/04/us/politics/trump-corporate-tax-cut.html>.
- I want to meet you!
 - ▶ O.H. Thursday 1:30-2:30pm in Littauer 207.
 - ▶ Lunches Thursday at 12pm via <https://c2t.fas.harvard.edu/> or at House/Annenberg: sign up link.

WHERE WE ARE IN THE COURSE

- ① First half: centered around growth models, **long-run** outcomes.
 - ▶ Real models: no money, flexible prices.
- ② Next lectures: **short-run** outcomes and stabilization policy.
 - ▶ Distinguish real and nominal interest rates and add sticky prices.
- ③ Big policy questions:
 - ▶ What does the central bank do?
 - ▶ Why do governments increase spending in recessions?
 - ▶ What explains the recent rise and fall in inflation?

NEXT BLOCK OF LECTURES

- 1 Monetary policy
- 2 Housing day
- 3 IS-MP
- 4 New Keynesian and ZLB
- 5 Inflation

MONETARY POLICY

Harvard Economics 1011B
Professor Gabriel Chodorow-Reich
Spring 2024

OUTLINE

- 1 MONEY
- 2 REAL AND NOMINAL INTEREST RATE
- 3 MONETARY POLICY
- 4 TERM STRUCTURE OF INTEREST RATES

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CHARACTERISTICS OF MONEY

- ① **Unit of account:** 🍏 costs \$1, 🍊 costs \$2,... Then $2 \text{ 🍏} = 1 \text{ 🍊}$. N goods characterized by N prices. Without unit of account, N goods requires $N(N+1)/2$ relative prices.
 - ▶ Refer to as *numeraire*.
- ② **Medium of exchange:** Trade 🍏 for dollars, dollars for 🍊. Solves “double coincidence of wants” in barter economy.
- ③ **Store of value:** Perfectly safe at short horizons.
- ④ **Fiscal backing:** Can it be used to pay taxes?
- ⑤ **Non-interest bearing:** Does holding it accrue interest?

MONEY?

Instrument	Unit of account	Medium of exchange	Store of value	Fiscal backing	Interest bearing
Currency	✓	✓	✓	✓	X
Checking deposits	✓	✓	✓	✓	X
Time deposits	✓	X	✓	X	✓
AAA rated security	X	X	✓	X	✓
Grocery store credit	X	✓	X	X	X
Bitcoin	✓	✓ (limited)	X	X	X

- Currency issuance directly controlled by government (inside money).
- AAA security, grocery store credit created by private agents (outside money).
- Fiscal backing $\Rightarrow$ unit of account, medium of exchange.
- Non-interest bearing, safe store of value for historical technological reasons.
- More: <https://mitpress.mit.edu/9780262533270/money-payments-and-liquidity/>.

Table 3

Not Seasonally Adjusted Components of M1 and Non-M1 M2. Billions of dollars.

Date	M1			Non-M1 M2	
	Currency ¹	Demand deposits ²	Other liquid deposits ³	Small-denomination time deposits ⁴	Retail money market funds ⁵
Sept. 2022	2,188.9	5,099.9	12,959.4	174.2	1,043.5
Oct. 2022	2,194.6	5,074.6	12,790.3	234.4	1,084.7
Nov. 2022	2,200.7	5,075.4	12,689.6	304.4	1,120.0
Dec. 2022	2,208.6	5,123.0	12,548.1	380.8	1,160.6
Jan. 2023	2,205.1	5,050.0	12,330.3	450.7	1,240.9
Feb. 2023	2,209.4	5,034.7	12,072.3	526.2	1,279.6
Mar. 2023	2,221.8	5,078.4	11,776.7	614.1	1,336.1
Apr. 2023	2,233.2	4,985.7	11,524.7	720.4	1,378.9
May 2023	2,246.1	4,974.2	11,271.4	804.4	1,415.0
June 2023	2,251.5	4,965.7	11,215.7	869.3	1,446.9
July 2023	2,247.7	4,940.3	11,146.5	916.1	1,473.5
Aug. 2023	2,243.1	4,896.9	11,086.6	959.0	1,509.4
Sept. 2023	2,239.4	4,885.1	11,008.5	991.4	1,544.6
Oct. 2023	2,237.6	4,923.0	10,861.6	1,017.2	1,596.7
Nov. 2023	2,241.1	4,928.7	10,852.7	1,049.9	1,650.3
Dec. 2023	2,247.8	5,032.6	10,842.7	1,075.2	1,691.7
Jan. 2024	2,244.6	4,972.4	10,798.6	1,092.8	1,754.2

<https://www.federalreserve.gov/releases/h6/20240227/>

CURRENCY

Value of currency in circulation, in billions of dollars as of December 31 of each year

	\$1	\$2	\$5	\$10	\$20	\$50	\$100	\$500 to \$10,000	TOTAL
2022	\$14.3	\$3.0	\$17.5	\$23.3	\$230.2	\$124.2	\$1,846.6	\$0.3	\$2,259.3
2021	\$14.0	\$2.8	\$17.1	\$23.2	\$237.9	\$122.9	\$1,769.3	\$0.3	\$2,187.5
2020	\$13.1	\$2.7	\$16.1	\$22.5	\$234.9	\$114.2	\$1,636.8	\$0.3	\$2,040.7
2019	\$12.7	\$2.7	\$15.8	\$20.9	\$190.5	\$92.3	\$1,424.7	\$0.3	\$1,759.8
2018	\$12.4	\$2.5	\$15.3	\$20.1	\$188.5	\$89.2	\$1,343.5	\$0.3	\$1,671.9
2017	\$12.1	\$2.4	\$14.8	\$19.6	\$183.8	\$86.4	\$1,251.7	\$0.3	\$1,571.1
2016	\$11.7	\$2.3	\$14.2	\$19.2	\$177.2	\$83.5	\$1,154.8	\$0.3	\$1,463.4
2015	\$11.4	\$2.3	\$13.7	\$19.0	\$171.3	\$79.8	\$1,082.2	\$0.3	\$1,380.0
2014	\$11.0	\$2.2	\$13.1	\$18.9	\$162.2	\$76.9	\$1,014.5	\$0.3	\$1,299.1
2013	\$10.6	\$2.1	\$12.7	\$18.5	\$155.0	\$74.5	\$924.7	\$0.3	\$1,198.3
2012	\$10.3	\$2.0	\$12.2	\$17.7	\$148.9	\$72.5	\$863.1	\$0.3	\$1,127.1
2011	\$10.0	\$1.9	\$11.8	\$17.2	\$141.1	\$69.6	\$782.6	\$0.3	\$1,034.5
2010	\$9.7	\$1.8	\$11.5	\$16.6	\$130.6	\$66.9	\$704.6	\$0.3	\$942.0
2009	\$9.6	\$1.7	\$11.2	\$16.2	\$127.5	\$65.3	\$656.4	\$0.3	\$888.3
2008	\$9.5	\$1.7	\$11.0	\$16.3	\$125.1	\$64.7	\$625.0	\$0.3	\$853.6
2007	\$9.3	\$1.6	\$10.8	\$16.2	\$121.8	\$63.0	\$569.3	\$0.3	\$792.2
2006	\$9.0	\$1.5	\$10.5	\$16.0	\$119.2	\$62.8	\$564.1	\$0.3	\$783.5
2005	\$8.8	\$1.5	\$10.3	\$15.5	\$115.4	\$62.1	\$545.0	\$0.3	\$758.8
2004	\$8.3	\$1.4	\$9.8	\$15.1	\$107.6	\$60.6	\$516.7	\$0.3	\$719.9
2003	\$8.2	\$1.3	\$9.7	\$15.1	\$107.8	\$59.9	\$487.8	\$0.3	\$690.2
2002	\$8.0	\$1.3	\$9.4	\$14.9	\$103.7	\$58.5	\$458.7	\$0.3	\$654.8

Includes Federal Reserve notes, U.S. notes, and currency no longer issued

https://www.federalreserve.gov/paymentsystems/coin_currircvalue.htm

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DEFINITION: NOMINAL INTEREST RATE

- $1 + i_{t,t+h}$: gross nominal return between t and $t + h$. That is, save/borrow \$1 in period t , receive/pay $\$1 + i_{t,t+h}$ in period $t + h$.
- Example: Certificate of deposit.
- Example: Treasury discount bill with face value \$100.
- Example: credit card borrowing rate.

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Marcus
by Goldman Sachs

Savings ^

CDs v

Investments v

Credit Cards v

Special Promotional CD Term

14-Month Term

5.40% APY ⓘ

Special Offer: Boost your savings with a special rate on our High-Yield CD. Lock in a fixed rate for a 14-month term with a guaranteed return on your money. Offer expires 4/17/24.

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TREASURY AUCTION RESULTS

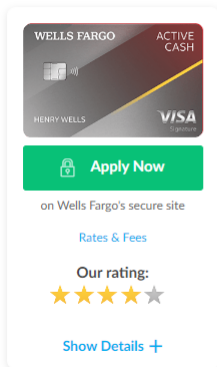
Term and Type of Security	364-Day Bill
CUSIP Number	912797JR9
High Rate ¹	4.570%
Allotted at High	99.43%
Price	95.379222
Investment Rate ²	4.814%
Median Rate ³	4.540%
Low Rate ⁴	4.480%
Issue Date	January 25, 2024
Maturity Date	January 23, 2025

- Example: credit card borrowing rate.

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- Example: Certificate of deposit.
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- Example: credit card borrowing rate.

Wells Fargo Active Cash® Card



Intro Balance Transfer APR: 0% intro APR for 15 months from account opening on qualifying balance transfers, then 20.24%, 25.24%, or 29.99% Variable APR after that intro period is up.

Intro Purchase APR: 0% intro APR for 15 months from account opening on purchases, then 20.24%, 25.24%, or 29.99% Variable APR.

Rewards: The rewards program for the Wells Fargo Active Cash® Card may look simple at first, but don't let that fool you – if you're a cardholder, you'll earn unlimited 2% cash rewards on purchases. 2% unlimited is unheard-of in the world of rewards credit cards so this card is absolutely not one to miss.

Extras: If you're a new cardholder, you'll earn a \$200 cash rewards bonus after spending \$500 in purchases in the first 3 months from account open.

Annual Fee: \$0.

EULER EQUATION REVISITED

- P_t : dollar price of a unit of consumption (of what?) in period t .
- $1 + \pi_{t,t+1} = P_{t+1}/P_t$: gross inflation rate between t and $t+1$.
- Consumption Euler equation:

$$\frac{1}{P_t} u'(C_t) = \beta \frac{(1 + i_{t,t+1})}{P_{t+1}} u'(C_{t+1}).$$

- Interpret: \$1 in period t buys $1/P_t$ units of consumption, which are valued at $u'(C_t)$. Or save \$1, receive $\$1 + i_{t,t+1}$ in $t+1$, purchase $(1 + i_{t,t+1})/P_{t+1}$ of consumption in $t+1$, valued at $\beta u'(C_{t+1})$.

FISHER EQUATION

- Rewrite consumption Euler equation:

$$u'(C_t) = \beta \frac{(1 + i_{t,t+1})}{\left(\frac{P_{t+1}}{P_t}\right)} u'(C_{t+1}) = \beta \frac{(1 + i_{t,t+1})}{(1 + \pi_{t,t+1})} u'(C_{t+1}).$$

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- Consumption Euler equation in terms of real interest rate $r_{t,t+1}$:

$$u'(C_t) = \beta(1 + r_{t,t+1})u'(C_{t+1}).$$

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- Approximation using $(1 + r_{t,t+1})(1 + \pi_{t,t+1}) \approx 1 + r_{t,t+1} + \pi_{t,t+1}$:

$$r_{t,t+1} = i_{t,t+1} - \pi_{t,t+1} \text{ (Fisher equation).}$$

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- Fisher equation with uncertainty over path of inflation:

$$r_{t,t+1} = i_{t,t+1} - E_t \pi_{t,t+1}.$$

RELATIONSHIP BETWEEN M AND i

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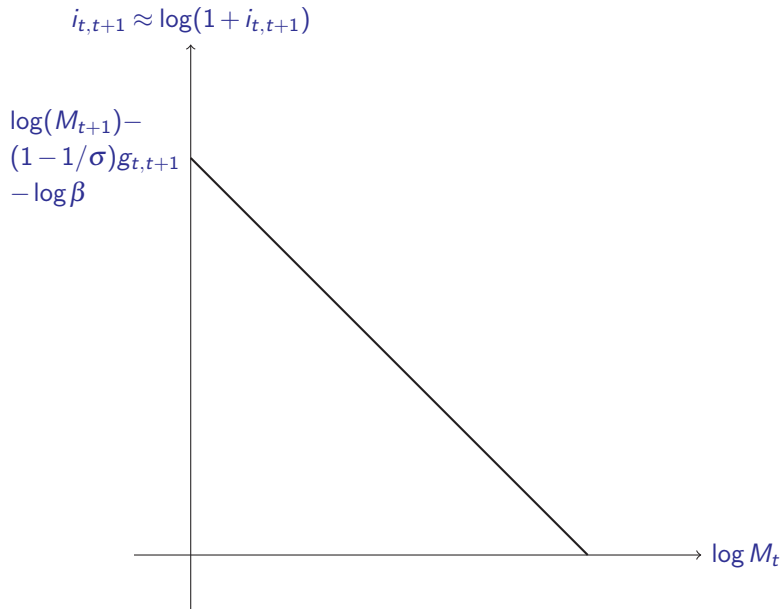
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 - ▶ Or fix M and let i adjust.
- Later in course: what happens if M fixed and implied i is negative?

LM CURVE



SUMMARY

- Real interest rate determines consumption today given (expected) future consumption: $u'(C_t) = \beta(1 + r_{t,t+1})u'(C_{t+1})$.
- Money supply and monetary policy determine nominal interest rate.
- Nominal and real interest rates linked through expected inflation.

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FEDERAL RESERVE DUAL MANDATE

- Federal Reserve Act as amended in 1977: *“The Board of Governors of the Federal Reserve System and the Federal Open Market Committee shall maintain long run growth of the monetary and credit aggregates commensurate with the economy’s long run potential to increase production, so as to promote effectively the goals of maximum employment, stable prices and moderate long-term interest rates.”*
- In practice, moderate long-term interest rates seen as bi-product of stable prices and maximum employment.

DUAL MANDATE IN PRACTICE: INFLATION TARGETING

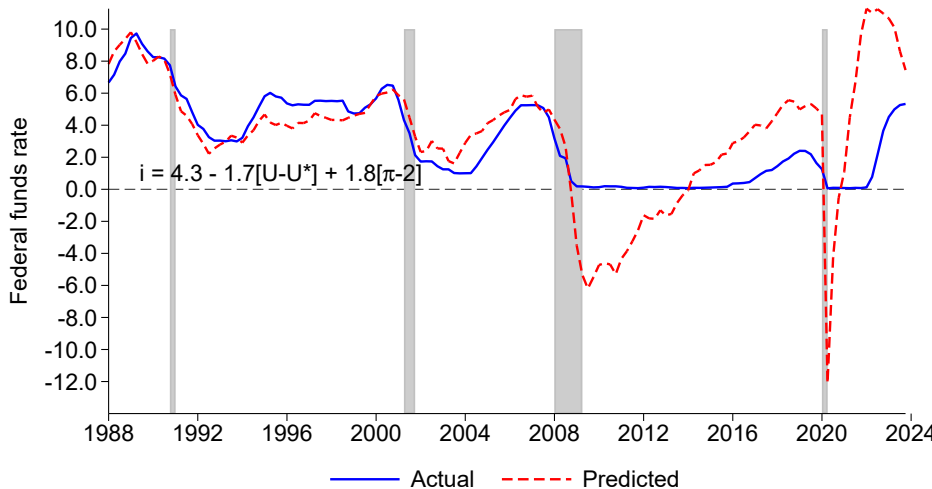
- Since 2012, Federal Reserve has practiced “inflation targeting.”
- Statement of longer run goals: *The inflation rate over the longer run is primarily determined by monetary policy, and hence the Committee has the ability to specify a longer-run goal for inflation. The Committee reaffirms its judgment that inflation at the rate of 2 percent, as measured by the annual change in the price index for personal consumption expenditures, is most consistent over the longer run with the Federal Reserve’s statutory mandate... In order to anchor longer-term inflation expectations at this level, the Committee seeks to achieve inflation that averages 2 percent over time... Monetary policy actions tend to influence economic activity, employment, and prices with a lag. In setting monetary policy, the Committee seeks over time to mitigate shortfalls of employment from the Committee’s assessment of its maximum level and deviations of inflation from its longer-run goal... Moreover, sustainably achieving maximum employment and price stability depends on a stable financial system.*

MONETARY POLICY FEEDBACK RULE (TAYLOR RULE)

$$i_{t,t+1} = r_{t,t+1}^* + \pi_{t,t+1}^* + \phi_Y(Y_t - Y_t^*) + \phi_\pi(\pi_{t,t+1} - \pi_{t,t+1}^*).$$

- Y_t^* : Fed's target for output; $\pi_{t,t+1}^*$: Fed's target for inflation (2%);
 $r_{t,t+1}^*$: real interest rate when $Y_t = Y_t^*$ and $\pi_{t,t+1} = \pi_{t,t+1}^*$.
- $\phi_Y > 0$: when output is below target, Fed lowers interest rates.
- $\phi_\pi > 0$: when inflation is above target, Fed raises interest rates.
- Assumption: ϕ_Y, ϕ_π large enough, and inflation sluggish enough, that Fed raises real interest rate. This is usually true.

ESTIMATED TAYLOR RULE



Fitted rule estimated over 1987:Q1-2007:Q4.

OTHER MONETARY REGIMES

- Money growth targeting.
- Price-level targeting.
- Nominal GDP targeting.
- Hybrid.
- Fixed exchange rate.
- Gold standard.
- Commodity standard.

i VERSUS *M* IN PRACTICE

- 1979: Paul Volcker appointed Federal Reserve Chairman.
- Needs to convince public he is serious about bringing down inflation.
- Emphasizes money and reserve growth targeting instead of *i*.

"All the News
That's Fit to Print"

The New York Times

LATE CITY EDITION

Weather: Partly sunny, breezy today; clear, cool tonight. Sunny, tomorrow. Temperature range: today 54-65; yesterday 53-67. Details on page 45.

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85 CENTS

President and Pontiff Issue a Plea At White House for World Peace

After Meeting Privately With Carter, John Paul Calls
for Arms Limitations — Big Mass on Mall Today

By FRANCIS X. CLINES

Special to The New York Times

WASHINGTON, Oct. 6 — Pope John Paul II was welcomed to the White House today by President Carter in a confluence of church and state in which both leaders pleaded to the world for peace.

Arriving on the last stop of his week-long tour of the United States, John Paul became the first Pope to enter the White House as he walked side-by-side with the President after a sun-frenched, smiling welcome before an assemblage of the Government's political and judicial leaders.

Proclaiming himself a "messenger of peace and brotherhood," the Pontiff said

that his purpose in coming to confer with the President was to "serve the cause of world peace, international understanding and the promotion of full respect for human rights everywhere."

After meeting privately with the President, the Pope issued a strong call for arms limitation. He did not cite the pending strategic arms limitation treaty, but his statement was immediately measured.

Remarks by the Pope are on page 37.

used as a possible lift for the treaty's chances of approval by the Senate. (Page 36.)

In greeting the leader of the world's Roman Catholics, the President, a born-again Christian, offered his first words in Polish: "Niech będzie Bog pochwałony — which to those of you who do not speak Polish, means: May God be praised."

Declaring the Pope for something the national spirit in his tour of six cities, President Carter declared, "The currents that flow below this brief ceremony reach into the very depths of individual lives and even to the breadth of relationships among sovereign nations."

The White House welcome, in which the Pope hugged Amy Carter as the Carters came to their doorstep to say hello, followed the beginning of still another day of motorcades past cheering crowds.

Streets Crowded and Festive

Millions of Americans have watched the Pope's progress about the nation this week, and today the streets of Washington were crowded and festive with still more people come to see the 58-year-old



President Carter applauding Pope John Paul II yesterday at the White House

Brezhnev Says Soviet Will Cut Forces in East Germany

By JOHN VINOCUR

Special to The New York Times

EAST BERLIN, Oct. 6 — Leonid I. Brezhnev said today that the Soviet Union will withdraw up to 20,000 troops and 1,000 tanks from East Germany during the next 12 months.

He said the Europeans need only to reject the deployment of the new American missiles, but to seek separate accommodation with Moscow. (Page 13.)

No Direct Mention of Cuba

Mr. Brezhnev, his words frequently

used by the Europeans, said that the new American missiles, but to seek separate accommodation with Moscow. (Page 13.)

In announcing the proposed reduction of forces in East Germany, he said: "In the course of the next 12 months, up to

Pope at Ease As Evangelist

He Shows Some Traits
Of American Preachers

By KENNETH L. BRIGGS

Special to The New York Times

CHICAGO, Oct. 6 — In the land where evangelists have thrived, Pope John Paul II has made his mark as a preacher. As he moved from stop to stop, dwelling at first on human rights, then concentrating on church discipline, his manner in some respects resembled the style of a traditional American revivalist.

Among the Pope's prime evangelistic traits is his ability to use the mood and

ANTI-INFLATION PLAN BY FEDERAL RESERVE INCREASES KEY RATE

DISCOUNT LEVEL A RECORD 12%

Package Will Also Alter Monetary
Policy and Bank Reserve Rule
— Interest Surge Likely

By STEVEN RATNER

Special to The New York Times

WASHINGTON, Oct. 6 — Paul A. Volcker, the chairman of the Federal Reserve Board, tonight announced a new package of measures, including higher interest rates, aimed at bringing inflation under control, calming financial markets and, indirectly, bolstering the dollar.

In a rare Saturday night news conference, Mr. Volcker said that the board of governors of the nation's central bank had voted unanimously to increase what is called the discount rate — the rate at which banks borrow from the Federal Reserve — by a two percentage point to a record 12 percent. The only other one-point increase in the Fed's history occurred as part of a dollar-rescue package last Nov. 1.

In addition, Mr. Volcker unveiled a new reserve requirement on certain bank borrowings that had helped promote an unhealthy expansion of bank credit, in the view of the Federal Reserve. Finally, the Fed plans to change the way it manages the money supply, with the prospect of more volatile but generally higher interest rates.

Further Brakes on Economy

In addition to higher interest rates, the new actions are likely to put further

MONETARY POLICY OPERATION

- The central bank (Federal Reserve, or “Fed”) sets i_t . How?
- Fed pays overnight interest on reserves to banks i_t^{ior} . No bank will lend at a rate below i_t^{ior} because Fed is safest counterparty.
- Fed borrows from non-banks at rate i_t^{rrp} (reverse repo). No non-bank will lend at a rate below i_t^{rrp} because Fed is safest counterparty.
- Ample reserves: banks hold excess reserves, ensuring i_t^{ior} is marginal safe lending rate.

OUTLINE

- 1 MONEY
- 2 REAL AND NOMINAL INTEREST RATE
- 3 MONETARY POLICY
- 4 TERM STRUCTURE OF INTEREST RATES

MONETARY POLICY AND LONG-TERM INTEREST RATES

- Fed controls short-term nominal rate. What about longer term rates?
- Definition: a pure discount nominal bond is a promise to pay a fixed sum (the face value) at a date in the future.
- Example: once every four weeks the U.S. Treasury issues a new 52 week Treasury bill which promises to pay \$100 in exactly 52 weeks.
- Example: most Mondays the Treasury issues a new 13 week bill.
- What is the relationship between the prices of 13 and 52 week bill?

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TREASURY AUCTION RESULTS

Term and Type of Security	364-Day Bill
CUSIP Number	912797JR9
High Rate ¹	4.570%
Allotted at High	99.43%
Price	95.379222
Investment Rate ²	4.814%
Median Rate ³	4.540%
Low Rate ⁴	4.480%
Issue Date	January 25, 2024
Maturity Date	January 23, 2025

- Example: most Mondays the Treasury issues a new 13 week bill.
- What is the relationship between the prices of 13 and 52 week bill?

BOND PRICES AND INTEREST RATES

- Suppose you pay P_t^{52} for 52 week Treasury bill.
- Compute return to holding bond:

$$R_{t,t+52} = \frac{100}{P_t^{52}}.$$

- Define nominal interest rate as $i_{t,t+52} = R_{t,t+52} - 1$.
- Approximation if $P_t^{52} \approx 100$:
$$i_{t,t+52} = \frac{100 - P_t^{52}}{P_t^{52}} = \frac{100 - P_t^{52}}{100} \frac{100}{P_t^{52}} \approx \frac{100 - P_t^{52}}{100} = 1 - .01 P_t^{52}.$$
- Example: bond costs \$99, interest rate is $100/99 - 1 = 0.010101\dots$
- Example: bond costs \$98, interest rate is $100/98 - 1 = 0.020408\dots$
- Result: bond prices move inversely with interest rates.

TERM STRUCTURE

- What is return on a 52 week Treasury bill?
- Consider alternative investment: buy 13 week bill and obtain return $R_{t,t+13}$, in 13 weeks invest proceeds in another 13 week bill, in another 13 weeks invest proceeds in another 13 week bill...
- Expected return:

$$t + 13 : R_{t,t+13}$$

$$t + 26 : E_t[R_{t,t+13}R_{t+13,t+26}]$$

$$t + 39 : E_t[R_{t,t+13}R_{t+13,t+26}R_{t+26,t+39}]$$

$$t + 52 : E_t[R_{t,t+13}R_{t+13,t+26}R_{t+26,t+39}R_{t+39,t+52}].$$

- This strategy is risky, because future interest rates are uncertain.
- Risk neutrality means indifferent between this strategy and investing directly in 52 week bill:

$$R_{t,t+52} = E_t[R_{t,t+13}R_{t+13,t+26}R_{t+26,t+39}R_{t+39,t+52}].$$

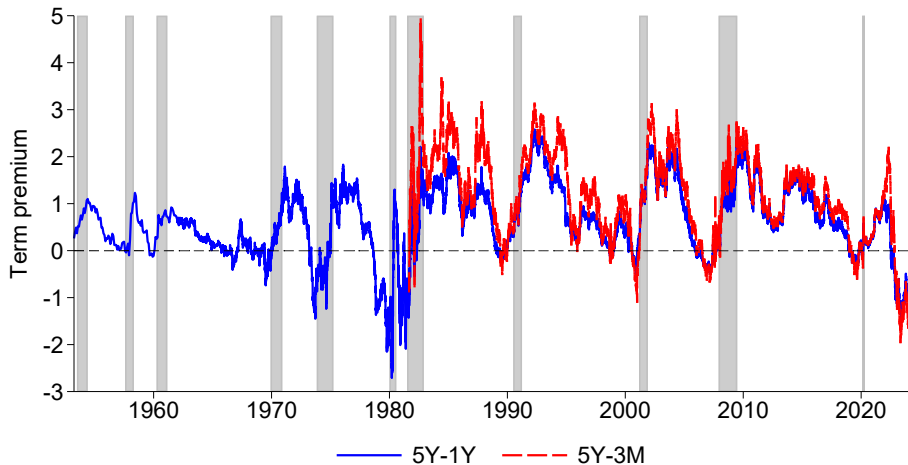
EXPECTATIONS HYPOTHESIS OF TERM STRUCTURE

- Useful fact: if x, y are small, then $(1+x)(1+y) \approx 1+x+y$.
- Using $R_{t,t+j} = 1 + i_{t,t+j}$ and the above fact:

$$i_{t,t+52} \approx E_t[i_{t,t+13} + i_{t+13,t+26} + i_{t+26,t+39} + i_{t+39,t+52}].$$

- This is the expectations hypothesis of the term structure. It states that a long-term interest rate is equal to the sum of expected short term interest rates over the contract horizon.
- Corollary: if the Fed can control short term interest rates, it can also affect long term interest rates through expectations of future short term interest rates.

YIELD CURVE



Why is inverted yield curve (negative term premium) recession warning?

EDITORS' PICK

The Inverted Yield Curve Continues To Flash A Recession Warning

Simon Moore Senior Contributor ①

I show you how to save and invest.

Follow



1

Feb 6, 2024, 06:15am EST

<https://www.forbes.com/sites/simonmoore/2024/02/06/the-inverted-yield-curve-continues-to-flash-a-recession-warning/>

ALTERNATIVE STORY FOR YIELD CURVE INVERSION

- Fed policy has been “restrictive” to bring down inflation.
- As inflation falls, policy will “normalize.”
- Chairman Powell (<https://www.federalreserve.gov/mediacenter/files/FOMCpresconf20240131.pdf>): *Our restrictive stance of monetary policy is putting downward pressure on economic activity and inflation... We believe that our policy rate is likely at its peak for this tightening cycle and that, if the economy evolves broadly as expected, it will likely be appropriate to begin dialing back policy restraint at some point this year.*

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